

Sandeep Sohal

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PROFESSIONAL SUMMARY

Results-driven Software Integration Engineer and Data Professional with 6+ years of experience designing and automating data pipelines, integrating enterprise software systems, and delivering ML-powered solutions. Proven ability to build end-to-end workflows using Python, SQL, and cloud platforms (AWS, Azure, GCP). Hands-on experience with REST APIs, PostgreSQL, MySQL, MongoDB, and Agile project delivery.

TECHNICAL SKILLS

Languages: Python, SQL, R, JavaScript, HTML/CSS, Java, Bash/PowerShell

Frameworks & APIs: Django, Django REST Framework, Flask, REST APIs, Git, GitHub

Data & Databases: PostgreSQL, MySQL, SQL Server, MongoDB, SQLite, Tableau, Power BI

ML & Data Science: PyTorch, TensorFlow, Librosa, pdfplumber, MediaPipe, RDKit, DeepChem, BERT, OpenCV

Cloud & DevOps: AWS (EC2, ECR, S3), Azure (Blob Storage), Google Cloud Platform, Docker, GitHub Actions, CI/CD

Tools & Platforms: Sage 300 ERP, KnowledgeSync/Sage Alerts, Agile/Scrum, Weka

EDUCATION

M.Sc. in Computer Science - Specialization in AI & Machine Learning 2020 - 2023

Toronto Metropolitan University (formerly Ryerson University) | CGPA: 3.96 / 4.33

Big Data Management Systems & Tools - Data Science Certification Jan. 2025 - Apr. 2025

University of Toronto, School of Continuing Studies

BBA Honours - Information Systems & E-Commerce 2015 - 2017

Trent University | Dean's Honour Roll

PROFESSIONAL EXPERIENCE

Software Integration Engineer & Database Administrator Feb. 2026 - Present

ADSS Global / Fitzpatrick Controllershship & Administrative Services (FCAS) | Toronto, ON

Python, PowerShell, SQL Server, PostgreSQL, REST APIs, Django, Git

- Designed and deployed an end-to-end document automation pipeline using Python and PowerShell, parsing, classifying, and routing structured data from multi-record PDF batches into date-partitioned output folders with automated CSV reporting.
- Built and maintained SQL Server and PostgreSQL database environments for enterprise ERP clients, including schema setup, module configuration, and ongoing database administration to support business workflows.
- Developed REST API integrations and file-based automation workflows connecting Python data processing scripts with PowerShell polling triggers, enabling reliable, event-driven pipeline execution across client systems.
- Coordinated email-based data distribution workflows using KnowledgeSync/Sage Alerts, scoping integration requirements and supporting handoff to downstream automation stages.
- Produced technical documentation, user requirement specifications, and customized client reports aligned to project scoping outlines; collaborated in an Agile team environment.
- Currently scoping and designing the system architecture for a machine learning-based customer churn prediction solution, including data pipeline design, model integration, and deployment planning for a client environment.

Academic Tutor

Oct. 2025 - Apr. 2026

Megamind Learning Centre | Mississauga, ON

Math, English, Science, Python

- Tutored students from Grade 5 through high school in Math, English, Science, and Python programming, tailoring lessons to individual learning needs and skill levels.
- Designed interactive, goal-oriented activities and assessments to track student progress, identify learning gaps, and improve academic performance.

Volunteer ML Engineer

Jun. 2025 - Aug. 2025

Squirrel Signs ASL | Toronto, ON

Python, SQL, GitHub, Azure, MediaPipe Holistic, CV

- Led a cross-functional team to architect an end-to-end real-time ASL-to-English translation MVP using MediaPipe Holistic, BiLSTM/Transformer sequence models, and a fine-tuned T5/BART language model.
- Managed dataset integration, preprocessing, and scalable data access via Azure Blob Storage using the 2M-FLORES-ASL dataset, enabling efficient model development.

Data Engineer

Mar. 2024 - Nov. 2024

Alenka Media Inc. | Vancouver, BC (Remote)

Python, SQL, GitHub, PyTorch, TensorFlow, Librosa, Agile

- Collaborated with a team of 5 to develop an audio classification model achieving 90% accuracy, categorizing audio by genre, tempo, and language using PyTorch and TensorFlow.
- Extracted and engineered 200,000+ audio features from MP3/M4A files using Librosa, generating spectrograms and MFCCs to build a high-quality training dataset.
- Reduced data preprocessing time by 40% by designing an optimized pipeline architecture for feature extraction and model ingestion.
- Managed project delivery using Agile methodologies, contributing to sprint planning, task tracking, and team code reviews via GitHub.

Research Scientist

Jul. 2024 - Nov. 2024

Konnifel - Indian Institute of Technology Madras (AI Lab) | Remote

Python, SQL, GitHub, RDKit, DeepChem

- Developed a chemical toxicity prediction system using Graph Convolutional Networks (GCNs) on a dataset of 500,000+ molecular structures preprocessed with RDKit and DeepChem.
- Web-scraped over 1 million chemical records from PubChem and MoleculeNet to expand dataset coverage for deep learning model training.

Marketing Data Analyst

Mar. 2020 - Nov. 2023

Health Medica | Kitchener, ON

Python, SQL, Weka, Hootsuite, GitHub

- Increased online engagement by 35% through data-driven digital marketing strategies backed by SQL-based segmentation and behavioral analysis.
- Forecasted sales trends with 85% accuracy using Weka data mining and SQL-based predictive models.

RESEARCH & PROJECTS

Skin Condition Detection Web App | *Python, OpenCV, CNN, PyTorch, TensorFlow, Flask, Docker, AWS, GitHub Actions*

Jan. 2025 - Mar. 2025

- Built a Flask web application to classify skin conditions using a CNN with transfer learning. Containerized with Docker, deployed on AWS (ECR & EC2), and integrated a full CI/CD pipeline via GitHub Actions.

Fake News Classification - NLP Research | *Python, TensorFlow, BERT, Google Colab GPU*

Dec. 2021 - Jan. 2022

- Processed 1M+ news articles for NLP training and achieved 94% test accuracy using a fine-tuned BERT model for binary classification of reliable vs. unreliable news sources.